

The digital backbone: ***Data center*** growth prospects in India

India | May 2025



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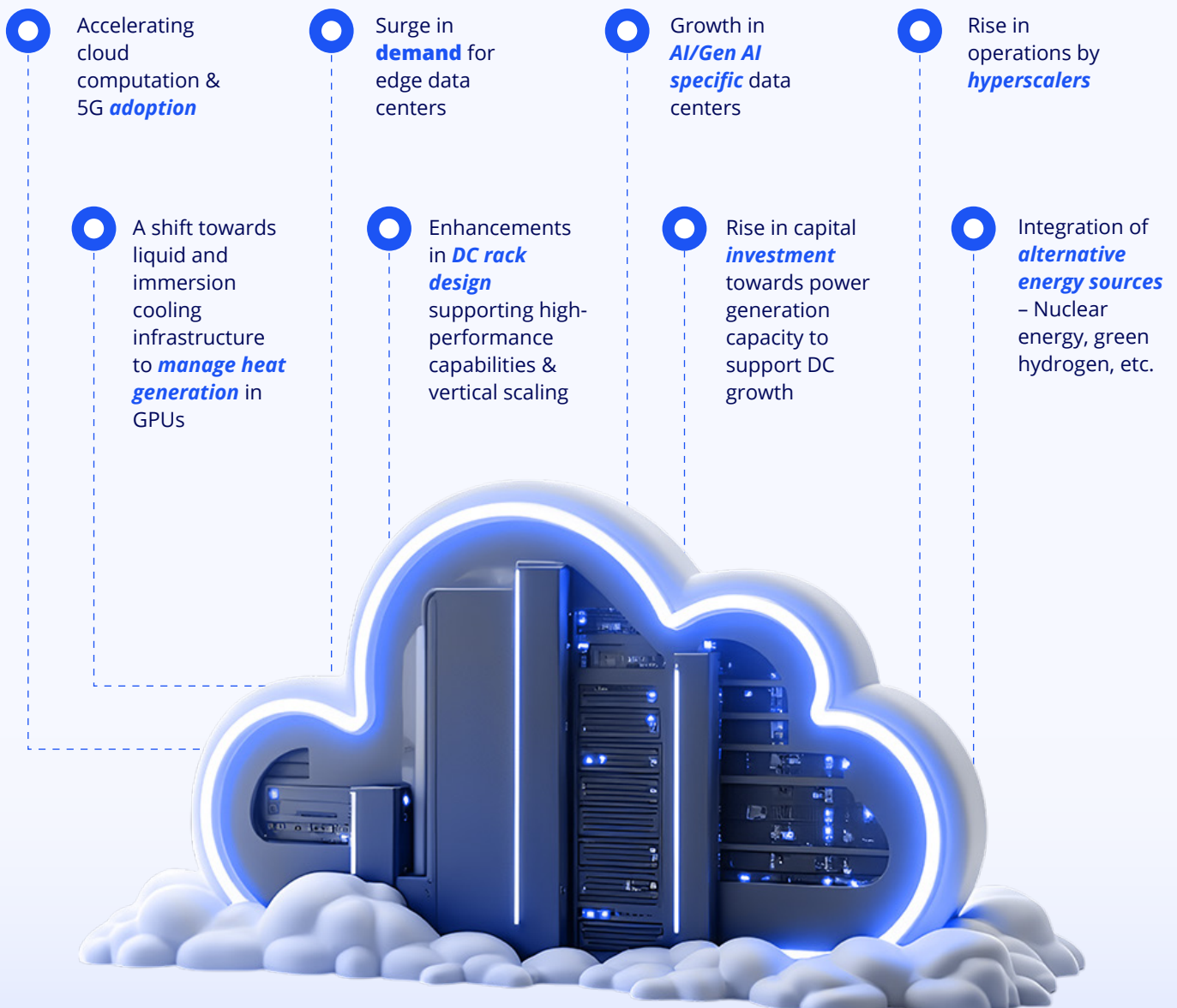
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Powering global digitalization

DC demand to soar amidst massive digitalization

Digitalization and disruptive technologies such as AI, cloud computing, machine learning, generative models etc. are taking rapid strides - generating, processing and storing large amount of data everyday. In 2024, global data collection touched close to 150 Zettabytes¹, a 2.3X times compared to 2020 levels. This is expected to further double up in the next 3-4 years driven by rise in internet penetration and increasing adoption of Everything as a Service (XaaS), Internet of Things (IoT), Big Data, AI etc. across various domains, starting from e-commerce to real estate, governance, defence, and beyond. As a result, there is a significant surge in demand for high-speed and secured data storage solutions. Industry experts project the global Data Center (DC) market to grow at a CAGR of 9-10% by 2030 up from the current ~USD 300 billion levels.² The coming years are likely to see phenomenal growth in the DC market, fueled by advancements in technology and increasing investments in related infrastructure. Leading operators are continuously innovating to enhance efficiency, security and scalability of data centers making them pivotal in driving economic growth across the world including India.

Key global DC trends



Key global statistics on digitalization



Global

~67%

of the global population (2024)



Internet penetration

Countries like USA and China have already started exploring & developing 6G technologies; Global 5G subscriptions will reach around 6.3 billion in 2030



5G/6G deployment

~19 GB per active smartphone (2024); likely to rise at a CAGR of 13% till 2030



Mobile data traffic per month

14.6%

in 2023



Technology & communication share in service exports

USD 180-200 billion; likely to rise at a CAGR of 28% in the next 5-6 years



AI market size



India

>55%

of India's population (2024)

Multiple countries including India are still deploying 5G technologies. 5G penetration in India is likely to have doubled during 2024 as compared to 2023

~30 GB per active smartphone (2024); likely to rise at a CAGR of 13% till 2030

48.2%

in 2023

USD >9 billion; likely to rise at a CAGR of 25-35% in the next 2-3 years

Source: Colliers, World Bank, Industry

Despite India having over 970 million internet users as of 2024, nearly three times that of the USA, its DC capacity remains significantly lower compared to mature and established markets like the USA, UK, Canada and Germany. Thus, by all means, India's DC industry is still nascent and holds immense potential for future growth, opening multiple opportunities for developers and operators. This growth potential is supported by established global connectivity through submarine cables, availability of land & power at comparatively lower costs, supportive government policies, and burgeoning demand. While the demand for large-scale conventional data centers will continue to grow in India, the demand for AI-driven, energy-efficient, and sustainability-focused data centers is set to soar significantly in the mid to long-term. Major global players and investors across the value chain have already started capitalizing on this transformative phase in the Indian DC industry to address the demand-supply mismatch.

OVERVIEW

Global data center market

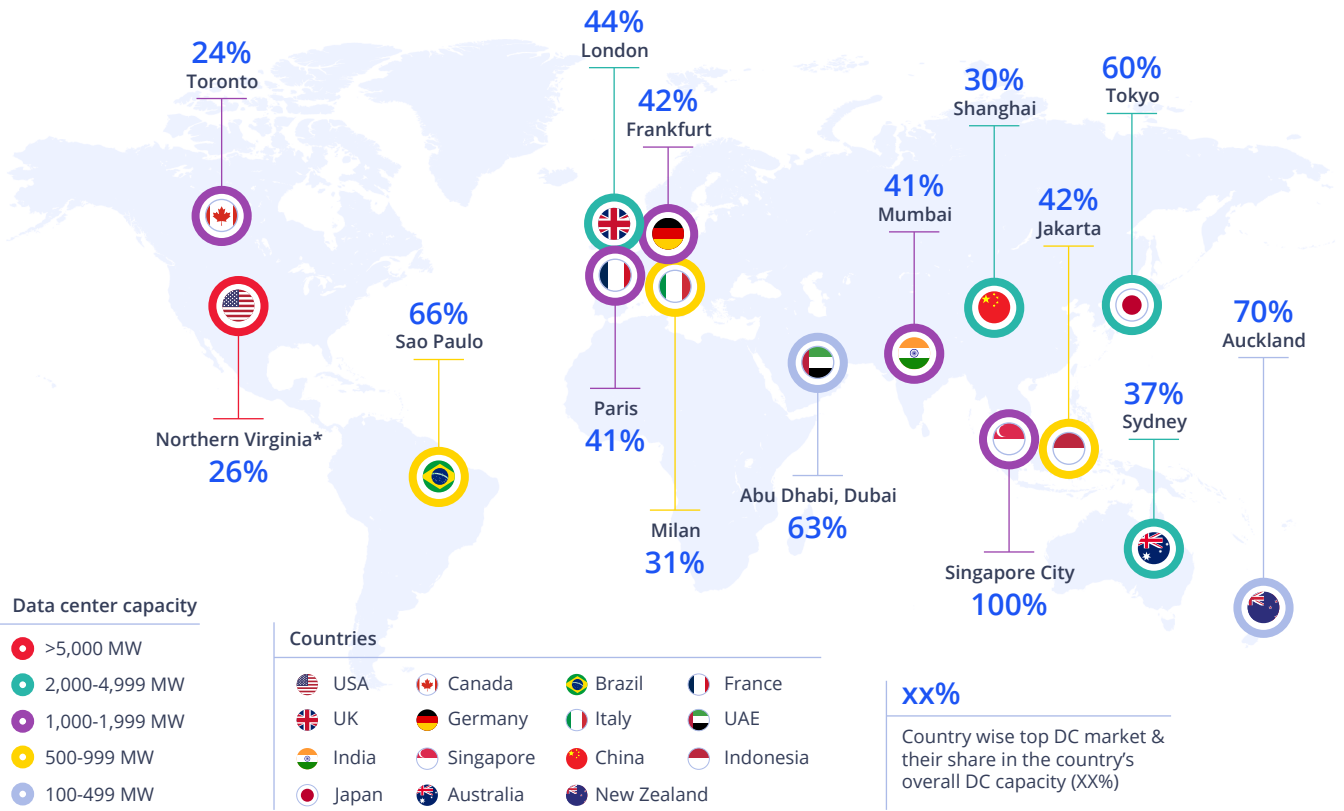
Maturity level of DC markets across the globe

	Mature Markets	Established Markets	Developing Markets
Country	USA, UK, Canada, Germany	Australia, China, Japan, Singapore, France	India, Italy, Indonesia, Brazil, New Zealand, UAE
Data center penetration	High	Medium	Low

Source: Colliers, Industry
Note: DC penetration is indicative of existing DC capacity in MW per 1 million of internet using population of respective country

The global DC industry has been witnessing robust growth and transformation across both established and emerging markets. While the USA, UK, Canada and Germany are mature DC markets, countries, like China, Japan, Australia etc. have a relatively established market and are prioritizing DC industry growth. Meanwhile, nascent and developing markets such as India, UAE, Indonesia, etc., with their robust government policies, improvements in infrastructure & connectivity, surge in technology adoption and land availability are attracting global investors and operators alike.

Prominent DC markets across the world



Source: Colliers, Industry
Note: Data center capacity represents total IT load capacity including occupied as well as unoccupied space | Data is as of April 2025 for India and as of December 2024 for other countries
* Northern Virginia (NoVA) includes counties of Arlington, Fairfax, Loudoun, and Prince William and cities such as of Alexandria, Falls Church, Fairfax, Manassas, and Manassas Park

Across countries, while established cities continue to drive the overall DC demand and supply dynamics, relatively smaller growth markets are gaining traction due to competitive pricing, government incentives, and improvements in DC infrastructure related to connectivity & power supply. These markets are increasingly complementing primary markets and are set to witness increased interest from both investors and operators alike. Some of the secondary DC markets with high growth potential globally include Austin, Columbus and Reno in the USA, Quebec City and Ontario in Canada, Manchester, Cardiff and Scotland in the UK, Guangzhou and Taipei in China, Chennai and Delhi NCR in India and Osaka in Japan.

India data center market

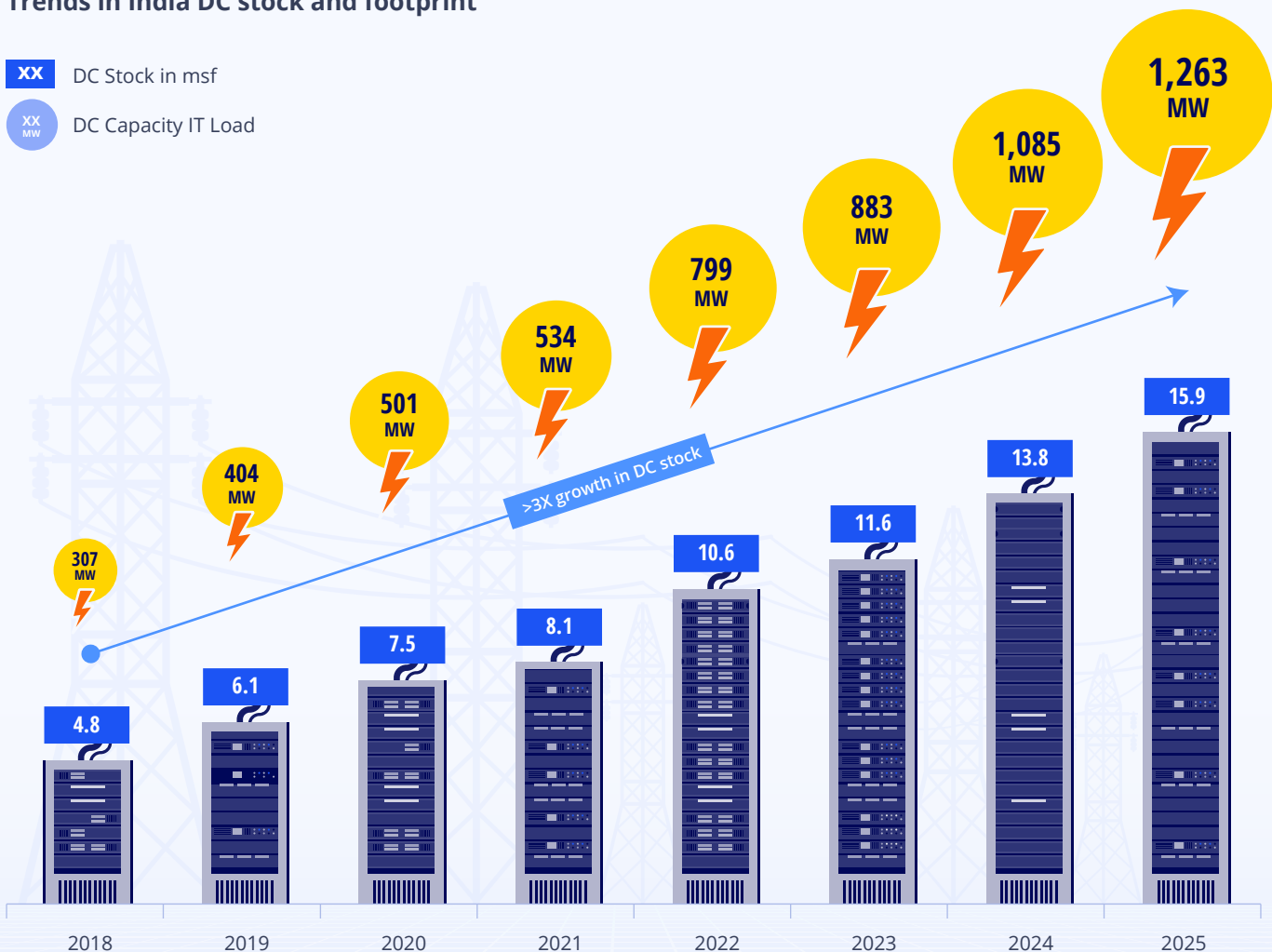
India's DC capacity more than doubled since 2020

India's DC market has entered a transformative phase and has scaled up significantly in recent years. The demand for colocation as well as hyperscale DCs has been gaining traction across major cities of the country. India's overall DC stock across top seven cities has increased over 3X times in the last 6-7 years to about 16 million sq ft, with about 1,263 MW capacity as of April 2025.

With rise in demand for digital & cloud services, increasing IoT & AI adoption, favorable government policies and higher internet penetration as well as data consumption, major DC operators both domestic and international are expanding their presence in the country, committing long-term investments in key markets including Tier II/III cities. India is rapidly becoming one of the fastest growing DC markets in the world driven by availability of land at affordable rates, submarine cable connectivity, uninterrupted power for commercial usage, and strict data localization norms.

In the coming years, the demand for high density racks, energy storage, and robust computation systems will further rise. India's DC market is primed for exponential growth and will have a competitive advantage in the wider APAC region considering inherent strengths in the four key components of DC industry development namely – land, electricity, infrastructure and skilled talent. Interestingly, high energy requirements will also expedite the need for green, energy-efficient data storage solutions and cooling systems in the Indian DC market.

Trends in India DC stock and footprint



Source: Colliers

*Data pertains to co-location data centers in top 7 cities - Bengaluru, Chennai, Delhi NCR, Hyderabad, Kolkata, Mumbai and Pune | Data center capacity represents total IT load capacity including occupied as well as unoccupied space | Data is as of April 2025

~16msf



Current DC stock*

1,263MW



Current DC Capacity*

2.5X



Rise in DC capacity since 2020



Share of top 5 DC operators

77%

India's DC market is highly concentrated amongst the top 5 DC operators who account for over 77% of the total DC capacity. However, mid-level DC enterprises are increasingly foraying into the expanding DC market of the country led by favorable ROIs and long-term growth prospects. Global players are likely to increasingly acquire or partner with these relatively newer enterprises and expand their India operations at a brisk pace over the course of next few years.

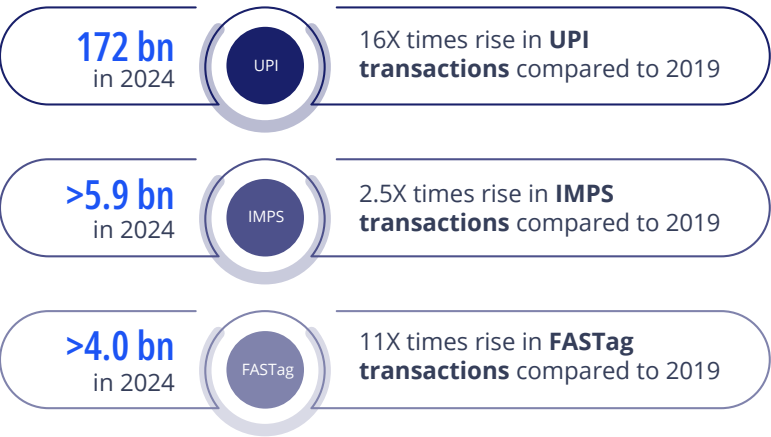
Source: Colliers

*Data pertains to co-location data centers in top 7 cities - Bengaluru, Chennai, Delhi NCR, Hyderabad, Kolkata, Mumbai and Pune | Data center capacity represents total IT load capacity including occupied as well as unoccupied space | Data is as of April 2025

Key growth drivers in India

Rapid digitalization to fuel DC expansion in India

Digitalization of Indian economy



Source: National Payments Corporation of India (NPCI)

India has undergone a significant digital transformation in the last decade. The development of robust digital services such as AADHAR (biometric identity), DIGI-Locker along with the introduction of cashless transactions through Unified Payments Interface (UPI), Immediate Payment Services (IMPS), and FASTag for seamless toll access has strengthened India's digital public infrastructure. India now ranks as the third largest digitalized economy in the world, after the USA & China and this rapid digitalization has amplified the DC demand in the country.

Favorable government policies

Digital India Mission 2015

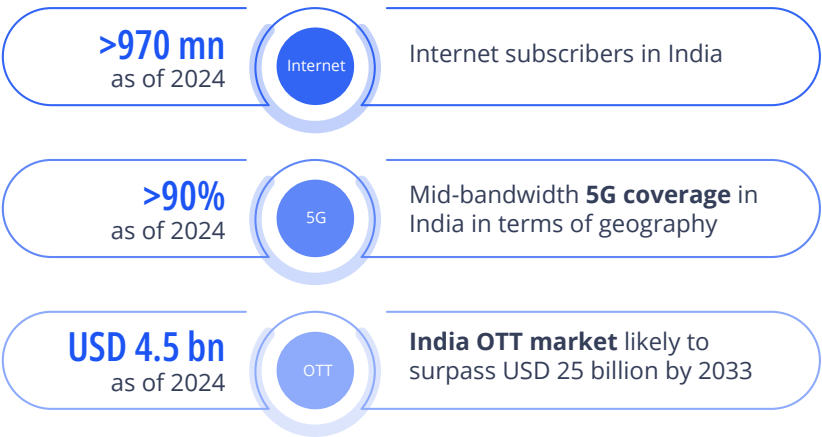
A flagship programme which aims to develop India into a digitally literate economic powerhouse. It focuses on improving online infrastructure and internet connectivity.

National Digital Communication Policy 2018

Aims to establish a robust and affordable digital communication ecosystem in the country by focusing on universal broadband connectivity, high-speed internet and attracting significant investments in IoT.

Rising data consumption & rapid 5G deployment

Quantum increase in data consumption and 5G service deployment will not only improve digital efficiency & governance but also benefit multiple sectors including manufacturing, retail, healthcare, agriculture and real estate. Moreover, increased penetration of social media and OTT platforms has led to exponential rise in data consumption, boosting the demand for DCs in the country.



Source: Telecom Regulatory Authority of India (TRAI), Industry

Favorable government policies

National E-commerce Policy 2019

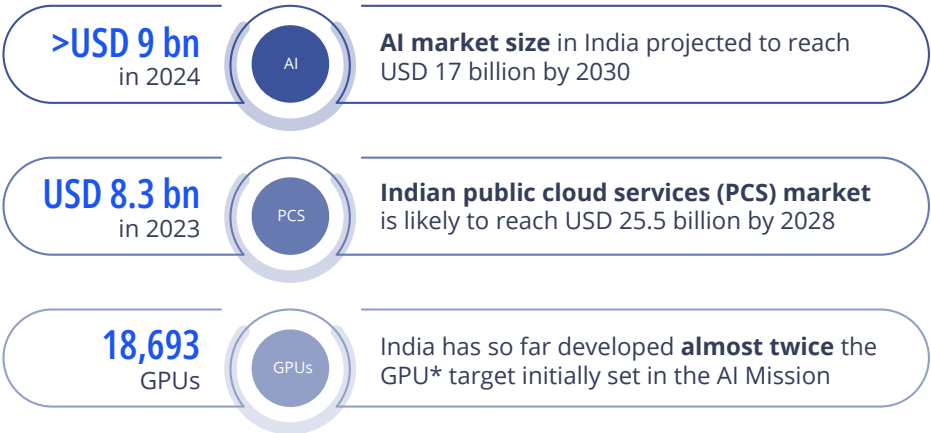
Aims to regulate e-commerce sector in India by advocating data protection and localization, safeguarding consumers' rights, fair competition and robust data governance.

National Broadband Mission (2.0) 2025

In continuation to the initial program launched in 2019, the mission aims to expand broadband infrastructure and affordable internet access across various satellite and remote areas of the country.

Growing convergence of cloud computation & AI

India is rapidly scaling up its public cloud service ecosystem and Artificial Intelligence (AI) market, backed by strong government initiatives such as the India-AI Mission and Public AI Computation, a growing AI start-up ecosystem and Generative AI-ready tech talent. This growing convergence of cloud-based services, AI and machine learning is likely to lead to multifold increase in DC demand over the next few years.



Source: NASSCOM, Ministry of Electronics and information Technology, Industry | *GPU- Graphics Processing Units

Favorable government policies

National Cloud Service Project 2014	India AI mission 2024
The project aims to accelerate delivery of e-services and optimize government ICT infrastructure while rolling out e-Gov applications. The initiative ensures optimal cloud utilization to speed up deployment of these applications.	The mission aims to promote ethical AI development and expand AI innovation by enhancing computing access, improving data quality, developing indigenous AI capabilities and attracting top AI talent.

Key government policies accelerating DC growth in India

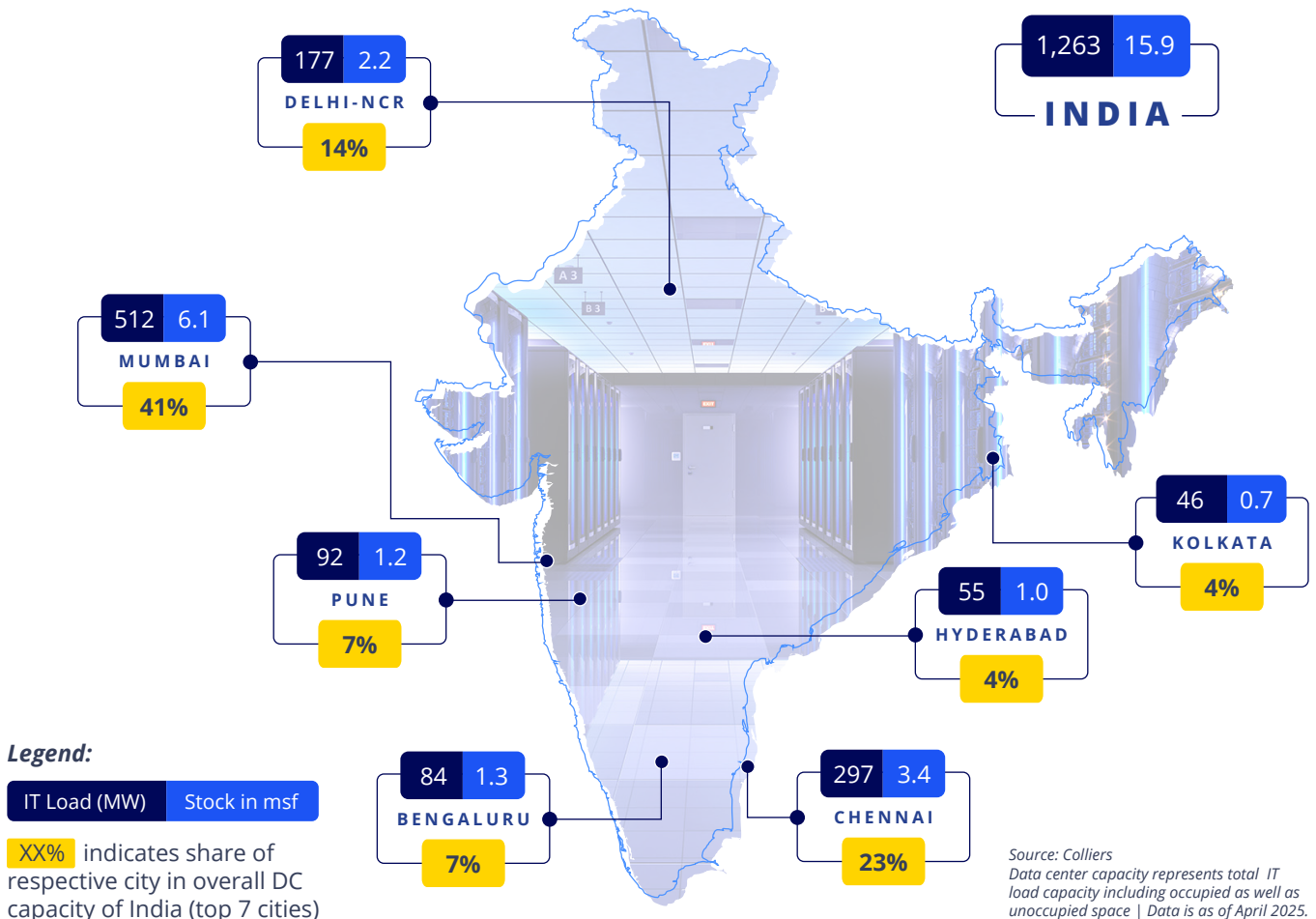
The expansion of DCs in India is supported by robust policies and regulatory frameworks at the central as well as state government level. The National Informatics Centre (NIC) has established National Data Centers (NDC) in cities like Delhi NCR, Pune, Hyderabad, Bhubaneswar and 37 smaller centers in multiple Tier II/III cities, providing essential cloud & disaster recovery services. These NDCs are equipped with high-speed connectivity and implement multiple e-governance projects across verticals.

Infrastructure status to DC industry	Draft Data Center Policy Framework 2020	Data Protection Bill 2023
The sector was accorded infrastructure status in 2022 under which DCs of more than 5 MW capacity have easier access to institutional credit at lower rates, thereby attracting foreign as well as domestic investments in the entire sector	The policy aims to make India a global DC hub through structural & regulatory interventions and establishing Data Center Economic Zones (DCEZs) to attract investments and strengthen domestic start-ups and MSMEs	The bill provides clear rules for handling data related to personal information, encouraging local storage and cross-border data management. It also ensures data accuracy, security, and timely deletion of data. This bill continues to boost the demand for data storage spaces, driving growth in the DC industry

City perspective

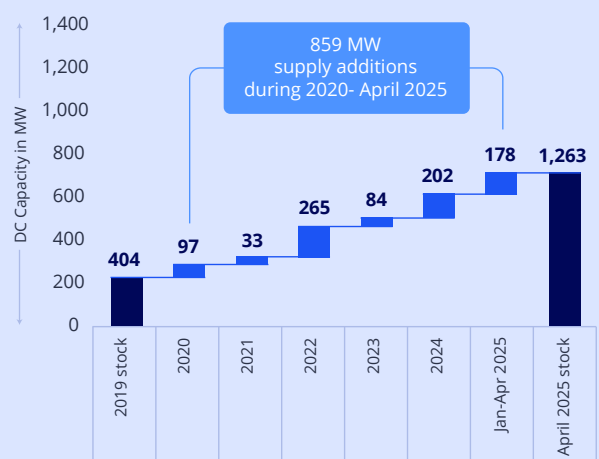
Mumbai accounts for 41% of the DC capacity across the top 7 cities

DC capacity & footprint across prominent Indian markets



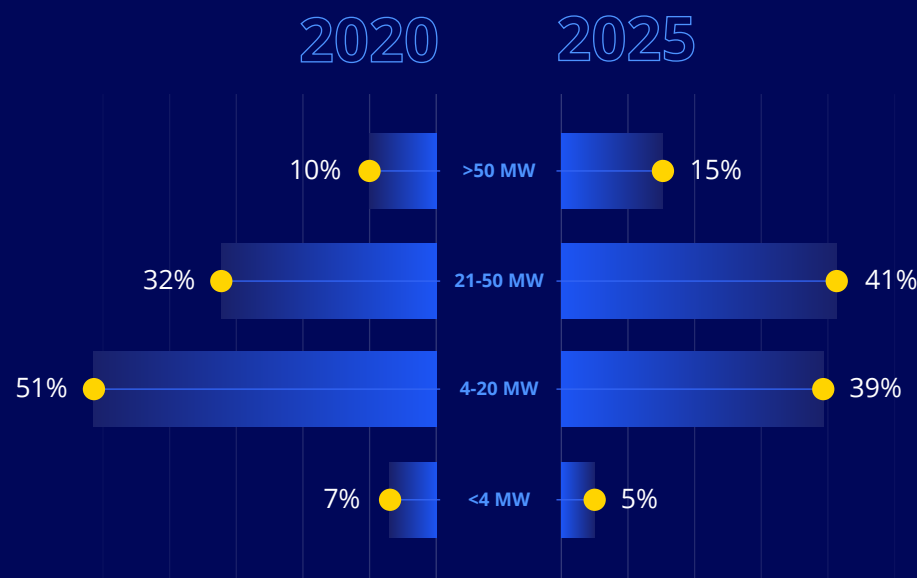
- Mumbai & Chennai together hold close to two-thirds of the DC capacity in India
- Mumbai continues to drive bulk of DC activity in India backed by robust power availability, infrastructure sufficiency and presence of multiple submarine landing stations
- While Chennai, Delhi-NCR, Pune and Bengaluru are already relatively established markets, Kolkata's potential of being a zonal data center hub for East India is noteworthy
- During 2024, majority of the top cities witnessed an uptick in data center supply addition
- Mumbai & Delhi NCR together saw about 111 MW of DC capacity additions in 2024 which accounted for 56% of new supply during the year

Trends in DC capacity/supply addition in MW across top 7 Indian cities



Source: Colliers
 Data center capacity represents total designed IT load capacity including occupied as well as unoccupied space. Data as of April 2025 | Top 7 cities include - Bengaluru, Chennai, Delhi NCR, Hyderabad, Kolkata, Mumbai, and Pune

Size-wise split of DC capacity across top 7 cities in India

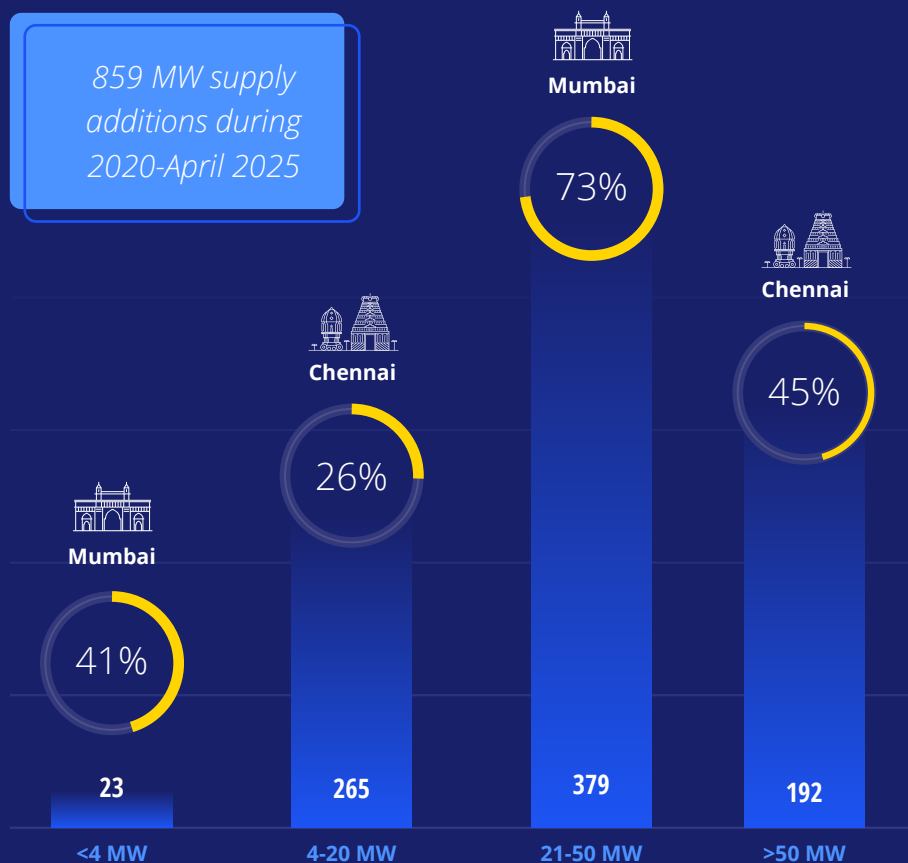


- As of April 2025, large-sized DCs (>20 MW) accounted for 56% of the overall capacity across the top 7 cities of the country, up from the 42% share in 2020.
- The rise in proportion of larger sized DCs is indicative of heightened traction in large hyperscale data centers, especially in the post-pandemic era

Source: Colliers

Data center capacity represents total designed IT load capacity including occupied as well as unoccupied space. Data as of April 2025 | Top 7 cities include – Bengaluru, Chennai, Delhi NCR, Hyderabad, Kolkata, Mumbai, and Pune

Size-wise split of DC supply addition (in MW) & city with maximum share (2020-April 2025)



- About 44% of the new supply during 2020- April 2025 were in 21-50 MW range. Meanwhile, supply infusion in >50 MW category accounted for a 22% share during the same period.
- Mumbai accounted for about three-fourths of the new supply additions in 21-50 MW category during 2020-April 2025 period followed by Chennai

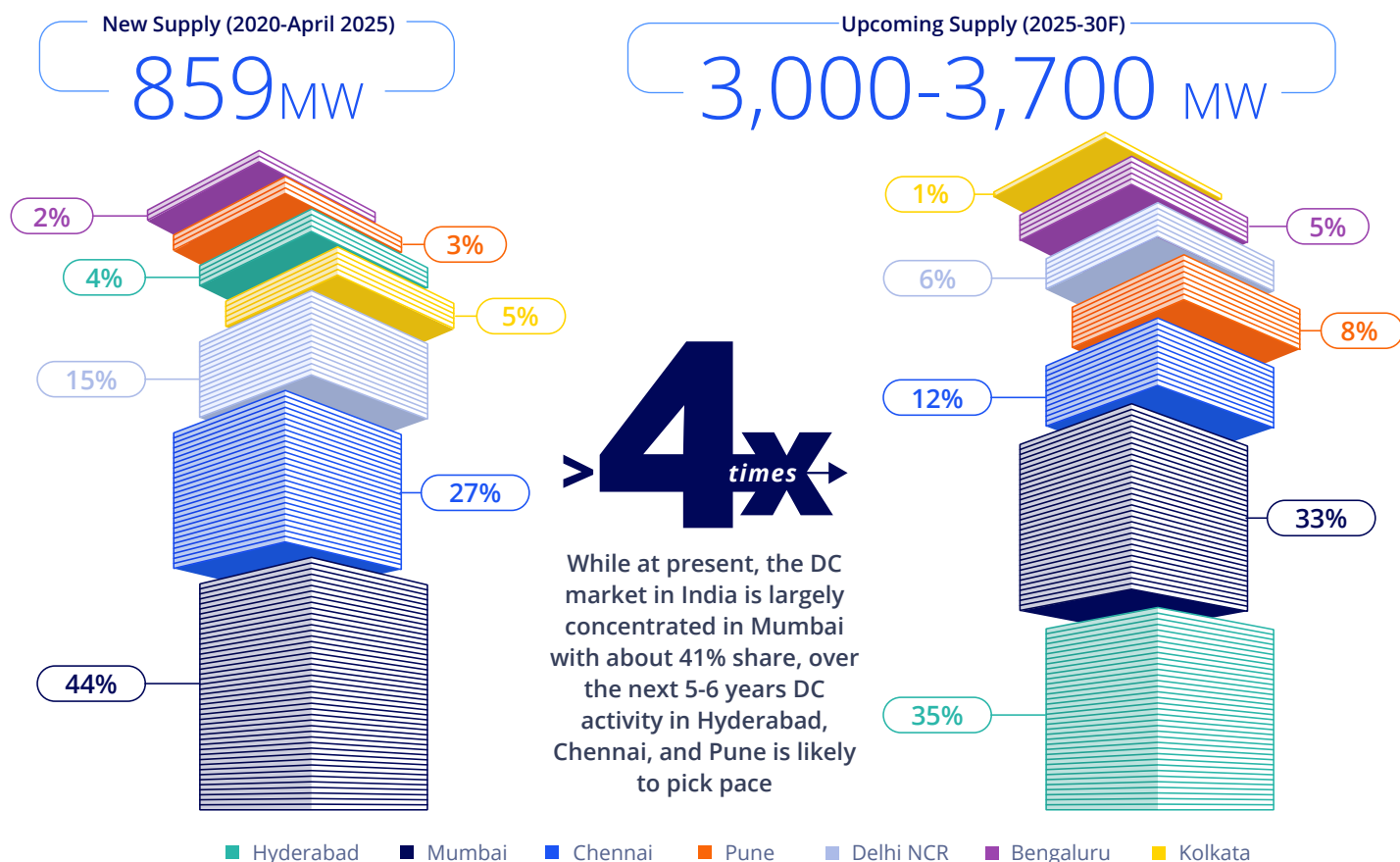
○ Indicates leading city in the particular IT load range along with its share across the top 7 DC markets of the country

Source: Colliers

Data center capacity represents total designed IT load capacity including occupied as well as unoccupied space. Data as of April 2025 | Top 7 cities include – Bengaluru, Chennai, Delhi NCR, Hyderabad, Kolkata, Mumbai, and Pune

Delineating DC market trajectory

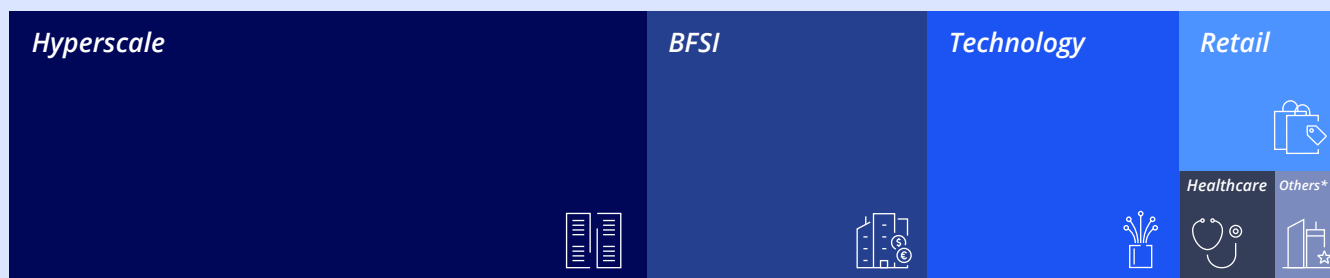
Supply side trends in Indian DC market



Source: Colliers | Data center capacity represents total designed IT load capacity including occupied as well as unoccupied space. Data as of April 2025

- 2022 and 2024 saw significant new supply in the Indian DC market. Almost 40% of the current DC capacity became operational in these two years
- During 2020-25, Mumbai witnessed the highest new supply with 44% share followed by Chennai & Delhi NCR at 27% and 15% respectively
- Chennai, Hyderabad, Pune and Kolkata have witnessed notable rise in new DC supply in the post pandemic era since 2022
- In the coming years, Mumbai will continue to dominate overall DC capacity share, witnessing a 3.4X times rise in its DC capacity by 2030 compared to the current levels
- Majority of the cities are set to witness significant influx of new supply in the next 5-6 years. Overall DC capacity in Hyderabad, Bengaluru and Pune, particularly is likely to see multifold growth compared to current levels
- A cumulative supply infusion of more than 600 MW in Chennai & Delhi NCR over the next 5-6 years is likely to result in more than 2X times capacity by 2030 compared to current levels

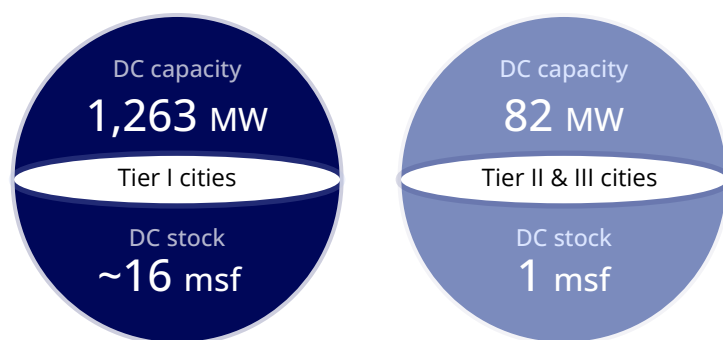
Key demand segments



Source: Colliers

Note: Key demand segments in the DC industry are aggregated based on market interactions and are indicative in nature | Others include OTT, Social media, etc. | Demand in Hyperscale DCs is largely driven by large cloud service providers and companies that process high-volume data etc.

DC landscape in Tier II/III cities



Source: Colliers
Data center capacity represents total designed IT load capacity including occupied as well as unoccupied space | Tier I cities include – Bengaluru, Chennai, Delhi NCR, Hyderabad, Kolkata, Mumbai, and Pune | Data as of April 2025

- India's DC industry is expanding beyond Tier I cities with operators and investors looking to gain first mover advantage in emerging Tier II/III cities. As of April 2025, Tier II/III cities such as Vijayawada, Jaipur, Mohali, Ahmedabad, Nashik, Kochi, etc. collectively accounted for 6% of the India's DC capacity to the tune of 82 MW. DC real estate footprint in these smaller cities is currently at around 1 million sq ft.
- Vijayawada accounted for the bulk of DC capacity amongst the Tier II/III cities followed by Mohali, and Jaipur.

DC capacity split in key Tier II/III cities



Source: Colliers
Data center capacity represents total designed IT load capacity including occupied as well as unoccupied space | Data as of April 2025 | Others* include – Bhubaneswar, Indore, Gandhinagar, Raipur, etc.

Going ahead, with increased adoption of cloud computation, AI & IoT, 5G deployment, rise of e-commerce in Tier II/III cities and steadfast implementation of state specific data center policies, DC growth is likely to accelerate in other emerging hotspots such as Indore, Raipur, Patna, Lucknow, Visakhapatnam, Hubballi, Coimbatore, Madurai, etc.

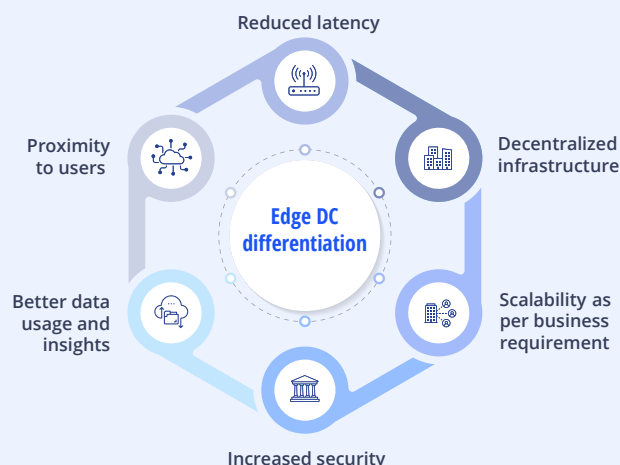
Edge data centers

- India's DC market is steadily expanding beyond large-scale colocation facilities and hyperscalers to edge data centers driven by increasing need for lower latency, real-time analysis, enhanced app performance, and business agility.
- An edge data center refers to a data center facility that brings computing power closer to where the data is generated or consumed. Distributing data processing and storage across various locations also improves security measures and protects against potential disruptions, ensuring seamless business operations.
- Hence, with MNCs, e-commerce players and technology companies expanding in Tier II/III cities, we expect the demand for edge DCs to amplify further in these smaller locations across the country.

Comparison of edge and hyperscale/ colocation DCs

	Hyperscale/ Colocation DCs	Edge DCs
Capacity	100KW-10MW and above	10-100KW
Rack density	5-20KW per rack	3-5KW per rack
Location	Centralized and often in high data consumption centers	Distributed and closer to end users
Latency	Moderate-High	Low

Source: Colliers, Industry



Investment landscape

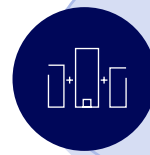
DC sector saw investment activity of about USD 14.7 bn since 2020

859 MW of DC capacity addition since 2020 has resulted in substantial investment activity to the tune of USD 14.7 billion in the last 5-6 years. DC investments were largely driven by the country's commitment towards digital transformation, supportive government policies and increasing demand for secured & robust data storage solutions. In the last 5-6 years, DC investment activity were majorly focused on land acquisition, project construction & development, etc. Of these investments, around 15% was for land related purposes. Overall investments in the DC sector is likely to rise further as operators & developers seek land-banking strategies and continue to scale-up their operations across markets including Tier II/III cities of the country. Global hyperscalers, too, have been ramping up their expansionary plans and developing built-to-suit (BTS) developments across key DCs.



USD 14.7 bn

Total investments in DCs (2020-April 2025)



15%

Share of investments towards land acquisition (2020-April 2025)

Note: The total investments include investments deployed during 2020-April 2025 period for development of data centers by PE players, DC operators and real estate developers.

*Source: Colliers
Data is as of April 2025*

Data centers accounted for 46% of the institutional inflows in alternative assets since 2020

DCs accounted for majority of the institutional inflows within alternative real estate assets, at USD 1.1 billion inflows since 2020. In recent years, global investors have been partnering with local operators to increasingly form joint ventures and capitalize on the growing DC market of the country. In fact, foreign inflows accounted for 86% of the investments in the DC sector since 2020, thus playing a dominant role in operator expansion and scaling-up of the market.

Institutional investments in DCs during 2020-25

USD 1.1 bn

Investments by institutional investors in DC during 2020-25

86%

Share of inflows by foreign institutional investors during 2020-25

~60%

Of the institutional inflows for DC during 2020-25 came in the last 2-3 years

61%

Mumbai & Chennai together attracted majority of the institutional inflows during 2020 - 25



*Source: Colliers
Data is as of March 2025*

What makes India's DC industry attractive to investors?



Rise in data storage requirements amidst significant internet & digital penetration through AI, big data, cloud computation and social media usage



Higher risk adjusted yields of about 16-17%, outperforming core assets (residential, office, warehouse) which typically offer less than 10% yields



Growing data storage demand from Healthcare, Engineering, E-commerce, media players, etc. apart from Hyperscalers, Technology and BFSI companies



Adoption of sustainable elements and energy-efficient measures to reduce the overall environmental impact of DCs



Cost arbitrage in land, labour and construction costs in contrast to other major global DC markets



Strong regulatory push and state-specific DC policies

Source: Colliers



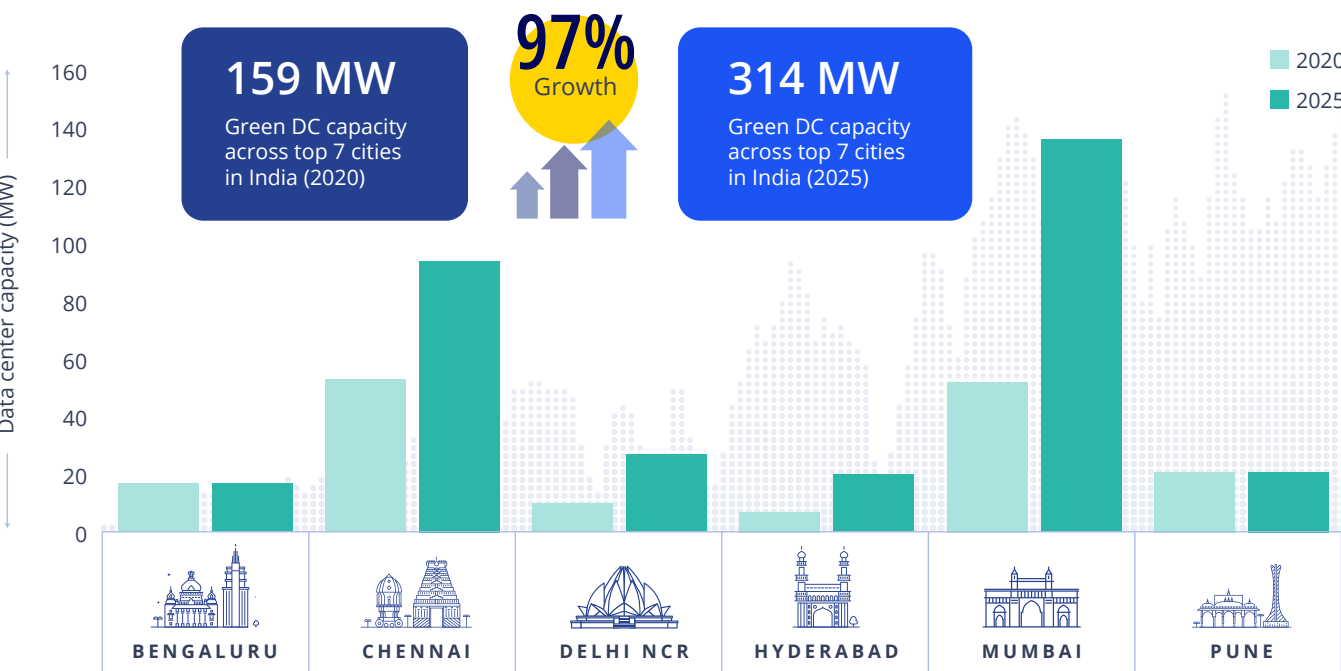
A progressive shift towards sustainability

India’s green DC capacity almost doubled since 2020

As data consumption continues to grow manifold, energy efficiency, sustainability and carbon emissions are likely to become focal considerations in the burgeoning DC footprint globally, including India. Data centers across the world are now actively focusing on achieving higher power and carbon efficiency, simultaneously targeting higher computing capabilities at lower costs. Sustainability in DC has now become one of the important parameters for operators, investors and occupiers as they continue to balance their business goals with global sustainability standards.

In the Indian context, as of April 2025, about one-fourth of the total DC capacity across major cities were green certified. In fact, green certified DC capacity in the country witnessed a CAGR of 15%, reaching from 159 MW in 2020 to 314 MW as of April 2025. At the city level, Mumbai and Chennai together accounted for close to three-fourths of the current green-certified DC capacity in India. In addition to the use of sustainable construction materials in DC development, usage of prefabricated modular components and retrofitting of existing DCs are likely to get a fillip in the coming years as operators look to improve operational efficiencies.

Green certified DC capacity



Green penetration in Indian DC market, indicated by share of green-certified DCs in the overall DC capacity stands at around **25%** (April 2025)

At 37%, **Hyderabad** has the highest green penetration, closely followed by Chennai at 32% (April 2025)

While **Mumbai** continues to dominate green-certified capacity, green DC capacity in Hyderabad and Delhi NCR have almost tripled since 2020

Source: Colliers
Note: Data Center capacity represents total designed IT load capacity including occupied as well as unoccupied space. Data as of April 2025 | Kolkata has minimal green DC stock

Leading DC operators in India are already focusing on imbibing sustainable practices and decarbonizing their portfolio – increasing their commitments towards achieving net zero targets by 2050. Apart from mitigating adverse environmental impact, green DCs provide enhanced rental premiums, asset valuation, brand recognition and reduced operational costs. Although leading operators have been able to reduce their carbon emissions by 25-50% by adopting sustainable strategies, a lot more needs to be done to achieve eventual net zero goals.

Design & development stage



Design stage

- Optimized & energy efficient designs integrating renewable energy resources & efficient cooling systems like liquid cooling, 2-phase immersion cooling, precision cooling etc.
- Using smart grids and virtualization allowing multiple servers to operate in a coordinated manner, reducing burnouts of physical servers and existing hardware
- Strategizing rack density and airflow by managing & reducing the mixing of hot and cold containments
- Altering facility design to enhance power, water and carbon usage effectiveness in line with globally acceptable sustainable standards

Development stage

- Using low emission construction materials and practices to promote sustainability at the construction stage itself
- Minimizing construction waste by focusing on measures aimed at reuse and recycle

Operational stage



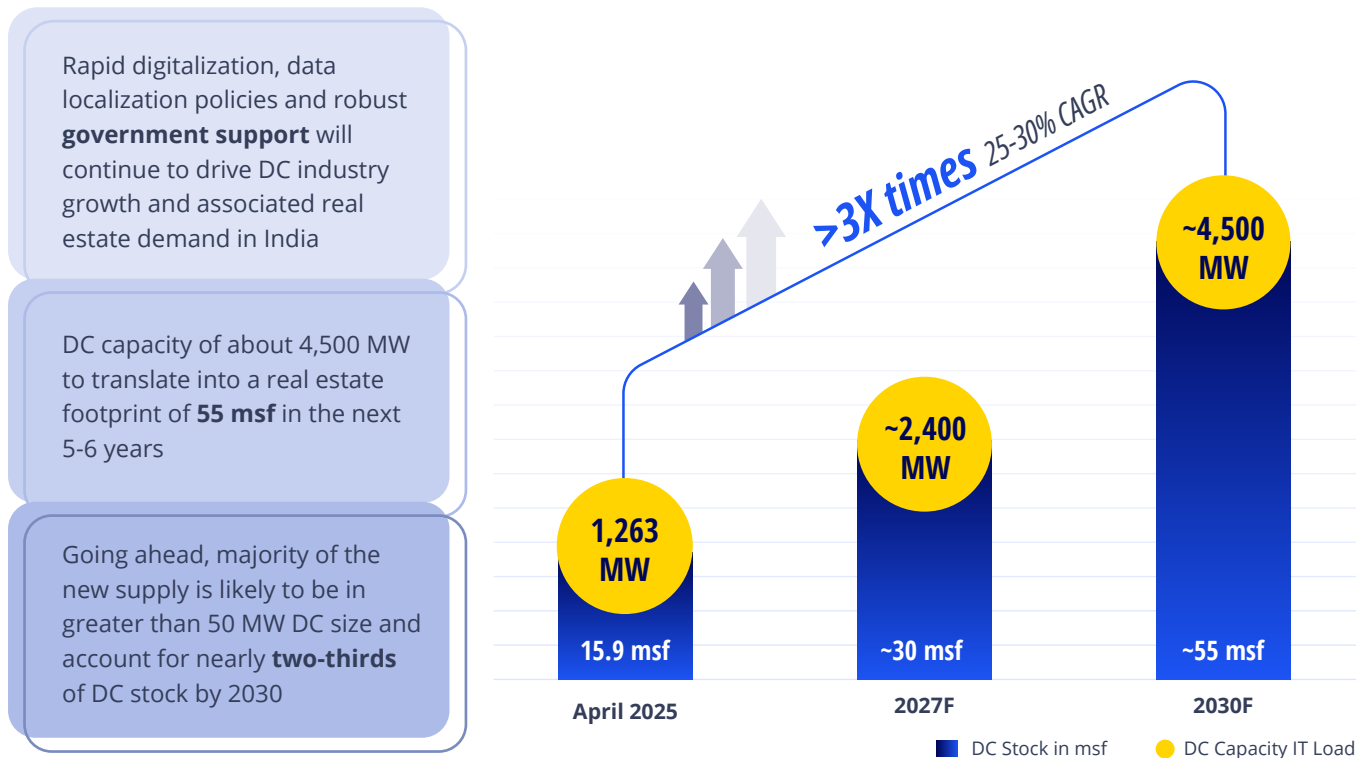
- Leveraging AI for energy management including predictive energy distribution, consumption levels and management of storage system workloads
- Incorporating ways to use waste heat to generate additional energy or redirecting the heat to other operations
- Embedding Data Center Infrastructure Management (DCIM) to provide a centralized platform for multiple functions and manage discrepancies in IT assets, cooling, storage etc.

Source: Colliers, Industry



Outlook

DC capacity to cross 4,500 MW by 2030 as USD 20-25 bn of future investments materialize



Source: Colliers

Note: Data Center capacity represents total designed IT load capacity including occupied as well as unoccupied space. | Data is as of April 2025

Size-wise split of DC capacity across top 7 cities in India (2030F)



Source: Colliers

Data Center capacity represents total designed IT load capacity including occupied as well as unoccupied space | Top 7 cities include – Bengaluru, Chennai, Delhi NCR, Hyderabad, Kolkata, Mumbai, and Pune | Share of <4 MW DCs in overall DC capacity is likely to become miniscule by 2030

Additional capital deployment of ~USD 20-25 billion in the next 5-6 years

- DCs are likely to attract investments to the tune of USD 20-25 billion amidst massive adoption of cloud computation and AI, in the next 5-6 years
- Real estate and construction cost typically account for about 30-40% of the overall DC development cost



Source: Colliers

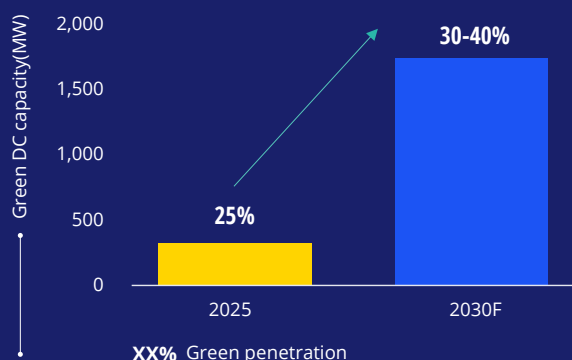
Green penetration to increase from 25% currently to 30-40% by 2030

- Leading DC operators will continue to focus on sustainability and increasingly opt for green-certifications. Hence, green certified DC capacity is likely to reach around 1,500-2,000 MW by 2030.
- Retrofitting of relatively older DCs can gain traction in next few years, driven by long-term benefits in terms of operational efficiency, average rentals and tenant quality

Source: Colliers

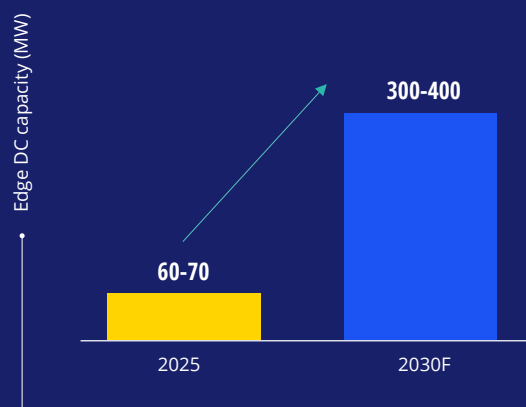
Note: Data Center capacity represents total designed IT load capacity including occupied as well as unoccupied space. | Data is as of April 2025

Green capacity and penetration forecast



Edge DCs to witness strong growth

Edge DC growth in the next 5-6 years



- The growing emphasis on edge computing and increasing 5G roll-out is likely to accelerate development of edge data centers. This will not only expedite DC growth in India but also lower average energy consumption and carbon emissions at the industry level.
- In addition to the growth envisaged in colocation DCs/Hyperscales, edge DC capacity is likely to increase at a CAGR of 35-40% in the next few years till 2030. Leading DC operators have already announced edge facility expansion in over 200 locations over the course of next 5-6 years.

Source: Colliers

Note: Data is as of April 2025

DC activity to rise in Tier II/III cities

- In the coming years emerging Tier II/III cities are likely to see a notable rise in both DC demand and supply amidst growing emphasis on low-latency infrastructure. DC capacity in Tier II/III cities is likely to touch 300-400 MW capacity in the next 5-6 years.
- The focus on regional development through state-specific policy interventions will drive expansion of digital infrastructure including DCs across the country.

Definitions

Classification based on size & power capacity

Colocation DC

A co-location DC is a large data center facility that rents out rack space to third-party clients for housing their servers or other network equipment.

Hyperscale DC

A hyperscale DC offers a portfolio of highly scalable applications and storage services to businesses. A hyperscale DC offers upwards of 5,000 servers with an aggregated space of at least 10,000 sq ft across floors and buildings within the same campus.

Edge DC

An edge data center is a small facility located close to the end users, designed to process and store data locally to reduce latency and improve performance.

Classification based on reliability & performance of DC infrastructure

Tier 1 is the simplest infrastructure, while Tier 4 is the most fault-tolerant and includes complex components. Each tier includes the required components of preceding tiers.

Tier 1: A Tier 1 DC has a single path for power & cooling and few backup components. It has an expected uptime of 99.671%.

Tier 2: A Tier 2 DC has partial backup for power & cooling systems, reducing the risk of downtime during maintenance. However, it is still susceptible to unexpected failures and has an expected uptime of 99.741%.

Tier 3: A Tier 3 DC has multiple paths for power & cooling and has systems in place ensuring higher availability & reliability. It ensures maintenance without taking entire DC system offline and has an expected uptime of 99.982%.

Tier 4: A Tier 4 DC is built to be completely fault tolerant, handling both planned and unplanned disruptions without affecting operations. It has an expected uptime of 99.995%.

Sources: Colliers International, Industry

Provisions for DC in select state policies

Categories	Category I	Category II	Category III	Category IV	Category V	Category VI
States	Fiscal benefits	Building norms, Power & connectivity infrastructure	Start-ups, MSMEs and EoDB	Skill development, labor & administrative support	Governance & implementation	Focus on sustainability
Karnataka						
Tamil Nadu						
Uttar Pradesh						
Odisha						
Telangana						
Maharashtra						
West Bengal						
Haryana						

Category I

- Capital grants
- Land subsidy/land conversion fee exemption
- Stamp duty exemption
- Interest subsidy/concession in registration charges
- Power tariff concession/electricity duty exemption

Category II

- Benefits related to FSI, parking, floor-to-ceiling height, rooftop use etc.
- Provision for power grid supply, bandwidth, internet, etc.
- DC specific land use provisions (Permission to set up data centers in commercial, industrial, and institutional zones)
- Provisions for water supply, water treatment, fuel backup, diesel generator stacking etc.

Category III

- Reforms specific to Ease of doing business (EoDB)
- Benefits for Start-ups/MSMEs

Category IV, V, VI

- Provisions provided (Yes)

Sources: Colliers, Industry

- Owing to the rising appeal of edge data centers in India, DC policies in states such as Uttar Pradesh, Tamil Nadu, West Bengal etc. have dedicated incentives for Edge DC
- Focus on green data center, energy efficiency and reduction of carbon footprint is also gaining momentum. As a result, many states are promoting sustainable practices in the DC industry by providing subsidies to green certified and energy efficient DCs.

Emerging DC markets in India



Sources: Colliers, Industry

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